Understanding CNN vs Vision Transformer: Source-Stage Results on CIFAR-10

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Abstract

This report summarizes the current source-stage experiments comparing a lightweight convolutional neural network (CNN) and a lightweight Vision Transformer (ViT) trained from scratch on CIFAR-10. The goal is to compare accuracy, data efficiency, robustness to occlusion and texture corruption, and class-level behavior before moving to downstream transfer learning. The main result is clear: in this setup, the CNN is substantially stronger on clean accuracy and low-data learning, while the ViT is markedly more robust to texture corruption. This pattern is consistent with prior literature showing that ViTs tend to require more data or stronger training recipes [3, 4], whereas CNNs often rely more heavily on local texture cues [5].

1 Introduction

Convolutional neural networks use locality and weight sharing as built-in inductive biases [1]. Vision Transformers instead tokenize images into patches and model global interactions with self-attention [3]. These differences often lead to different learning behavior: CNNs can work well in small-data settings, while ViTs typically benefit from larger-scale data or stronger pretraining and regularization [3, 4]. Separately, prior work has shown that standard CNNs can be strongly texture-biased [5].

This report tests whether these broader trends appear in a controlled from-scratch experiment on CIFAR-10 [2]. The present document covers only the source-stage study. Transfer learning to EuroSAT will be added after this stage.

2 Methodology

2.1 Task and comparison

Both models were trained for image classification on CIFAR-10. The comparison was designed to answer four questions:

- Which model is more accurate on the clean test set?
- Which model is more data-efficient?
- Which model is more robust to occlusion and texture disruption?
- What kinds of class confusions remain after training?

2.2 Models

The CNN baseline is a compact three-stage architecture with convolution, batch normalization, ReLU, max pooling, global average pooling, dropout, and a linear classifier. The ViT baseline is a lightweight patch-based transformer with patch embedding, learnable positional embeddings, a classification token, and transformer encoder blocks [3]. The goal was not to compare the strongest possible CNN against the strongest possible ViT, but to compare two lightweight models under the same training budget.

2.3 Interpretability

The pipeline also generates qualitative explanations for the full-data models. Grad-CAM is used for the CNN [6], and attention rollout is used for the ViT [7]. These artifacts are available, but the present report focuses on the quantitative source-stage results.

3 Experimental Setup

3.1 Dataset and preprocessing

CIFAR-10 contains ten object classes and uses RGB images of size $32 \times 32 \times 3$ [2]. The original training split has 50,000 images and the test split has 10,000 images. In this project, 5,000 images were held out from the original training set for validation, leaving a training pool of 45,000 images.

Training preprocessing used:

- `RandomCrop(32, padding=4)`
- `RandomHorizontalFlip()`
- `ToTensor()`
- channel-wise normalization with CIFAR-10 statistics

Validation and test preprocessing used only tensor conversion and normalization. This keeps evaluation deterministic.

3.2 Data-efficiency protocol

Separate runs were trained from scratch using 10%, 25%, 50%, and 100% of the 45,000-image training pool. This corresponds to 4,500, 11,250, 22,500, and 45,000 training images respectively. These were independent runs, not progressive continuation runs.

3.3 Robustness protocol

Robustness was evaluated using the 100% models only. Two test-time distribution shifts were applied:

- **Occlusion**: a square patch is masked.
- **Texture modification**: local patches are shuffled and perturbed with noise while preserving the global image layout.

3.4 Training details

All reported CIFAR experiments were trained for 100 epochs with AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch size 128, and seed 42. All values in this report come from one controlled split and one random seed.

4 Results

4.1 Clean accuracy and efficiency

Table 1 shows the main full-data comparison. The CNN reached 85.7% test accuracy, while the ViT reached 73.36%. The CNN therefore outperformed the ViT by 12.34 percentage points while using far fewer trainable parameters (0.37M vs 1.80M). It also trained slightly faster.

This result indicates that, in the current from-scratch CIFAR-10 setting, the CNN is the stronger source-stage model. The weaker ViT performance is consistent with prior work showing that transformers in vision tend to need larger-scale pretraining or more specialized data-efficient training recipes [3, 4].

Table 1: Full-data comparison on CIFAR-10. Macro-F1 values come from checkpoint-based evaluation.

Model	Params	Clean Acc.	Macro-F1	Occluded Acc.	Texture Acc.	Train Time
CNN	374,282	85.70%	0.858	80.32%	32.98%	13m 56s
ViT	1,803,850	73.36%	0.733	68.81%	50.26%	15m 46s

4.2 Data efficiency

The data-efficiency trend is consistent across all fractions. The CNN outperformed the ViT at every training budget:

Table 2: Test accuracy across training fractions.

Model	10%	25%	50%	100%
CNN	65.70%	75.08%	81.48%	85.70%
ViT	53.53%	60.27%	67.03%	73.36%

The CNN had a strong early gain from 10% to 25% data (+9.38 points), then continued to improve as more data was added. The ViT also improved steadily, but remained behind at every fraction. The gap between the two models stayed large: 12.17 points at 10% data, 14.81 points at 25%, 14.45 points at 50%, and 12.34 points at 100%.

This confirms that the CNN is more data-efficient in this setting. The ViT did not close the gap even after 100 epochs, which supports the interpretation that the current ViT setup still wants more data or stronger optimization support.

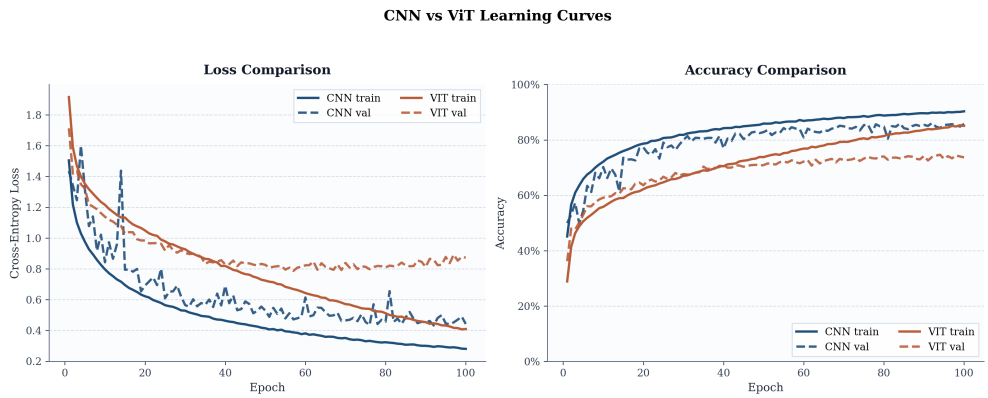
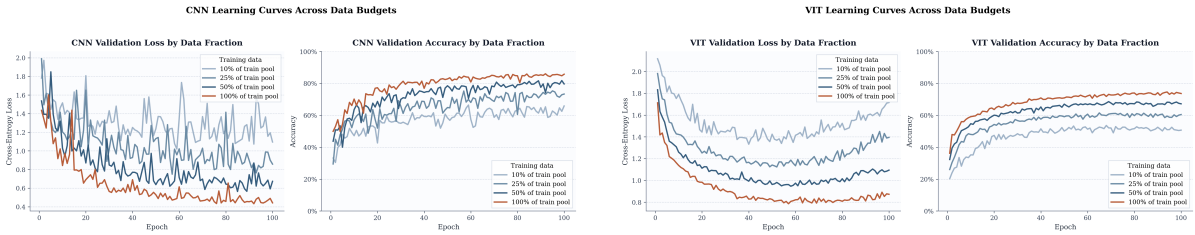


Figure 1: Shared learning curves for the full-data CNN and ViT runs. The CNN converges faster and to a higher validation accuracy.



(a) CNN across data fractions

(b) ViT across data fractions

Figure 2: Validation learning curves across data budgets. The CNN remains ahead throughout training, especially in the low-data regime.

4.3 Robustness to occlusion and texture modification

The robustness results are more nuanced than the clean-accuracy results.

Under occlusion, both models degraded moderately:

- CNN: 85.70% → 80.32% (drop of 5.38 points)
- ViT: 73.36% → 68.81% (drop of 4.55 points)

The ViT had a slightly smaller drop, but the CNN still retained much higher absolute accuracy.

Under texture modification, the difference was much larger:

- CNN: 85.70% → 32.98% (drop of 52.72 points)
- ViT: 73.36% → 50.26% (drop of 23.10 points)

This is the strongest result in the current study. The CNN collapses under texture disruption, while the ViT degrades much less. This pattern is consistent with the texture-bias literature on CNNs [5]. In other words, the CNN is better on standard accuracy, but the ViT appears to rely less heavily on local texture statistics.

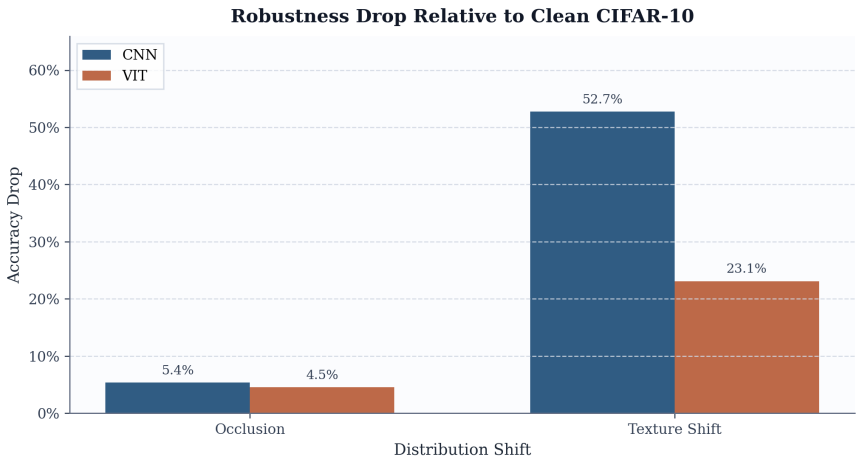


Figure 3: Accuracy drop relative to the clean CIFAR-10 test set. Texture corruption harms the CNN much more than the ViT.

4.4 Class-level behavior

Checkpoint-based evaluation shows that both models make their largest errors on visually similar animal classes. For the full-data runs:

- The CNN's lowest recalls were on *dog* (0.756), *bird* (0.777), and *cat* (0.806).
- The ViT's lowest recalls were on *cat* (0.569), *dog* (0.631), and *bird* (0.673).
- Both models performed best on vehicle classes, especially *automobile* and *ship*.

The dominant confusion pairs also support this interpretation. For the full-data CNN, the largest confusion was *dog* → *cat* (156 cases). For the full-data ViT, the largest confusion was also *dog* → *cat* (196 cases), with a similarly strong reverse confusion *cat* → *dog* (168 cases). These class-level errors suggest that the ViT's main weakness in this experiment is fine-grained discrimination among similar animal categories.

5 Conclusion

At the current source stage, the CNN is the better overall classifier on CIFAR-10. It is more accurate, more data-efficient, smaller, and slightly faster. However, the ViT shows a clear advantage under texture corruption, which is important because it suggests a different representation strategy rather than simply a weaker model.

The direct conclusion is therefore not "CNNs are better than ViTs." The more precise conclusion is:

- CNN is better for this from-scratch small-image classification setup.
- ViT appears less texture-dependent and therefore more robust to the specific texture-shift test used here.

This gives a strong foundation for the next stage of the project. The saved 100% CIFAR checkpoints can now be used for transfer learning on EuroSAT, where the key question will be whether the different source-stage behaviors translate into different downstream transfer performance.

References

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